Foundations of Computational Theology: Set-Theoretic Meta-Ethics and the Gödelian Completion

Rafael D. De Paz
Sovereign Architect
me@rdepaz.com

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Abstract

We introduce *Computational Theology*—a formal framework that applies the tools of mathematical logic, set theory, and control theory to classical theological questions. We present three foundational results. First, we define moral truth as a membership function in a master specification Ω_{root}, establishing ethics as a geometric property of a universal state-machine: $f(A_i) = 1 \iff A_i \subseteq \Omega_{\text{root}}$. Second, using Gödel's Incompleteness Theorems, we prove that any sufficiently complex self-contained system requires an *external root authority* to resolve undecidable propositions—formalizing the necessity of a transcendent ground. Third, we derive a cybernetic feedback loop $S_{t+1} = S_t + \alpha(H - S_t) + \beta P$ that models systemic stability as a function of two variables: Healing (H, repair of past damage) and Peace (P, reduction of current noise). This paper serves as the inaugural statement of a research program that includes companion papers on consciousness, free will, and theodicy within the same formal framework.

Keywords: computational theology, meta-ethics, Gödel's incompleteness, set theory, control theory, cybernetics, morality.

1 Introduction

These foundational principles operate on established invariants Bostrom [2003], Chaitin [1975].

The relationship between formal logic and theology has a long history, from Anselm's ontological argument (1078) through Gödel's ontological proof [Gödel, 1995] to Plantinga's modal logic reformulations [Plantinga, 1974]. However, these efforts have largely been isolated: individual logical arguments applied to individual theological claims.

We propose a unified framework—*Computational Theology*—that treats theology as a *formal system* subject to the same constraints as any other: consistency, completeness, decidability, and entropy. This is not a metaphor. We claim that theological propositions can be formalized, that ethical truths have geometric structure, and that the necessity of a transcendent ground can be *derived* from Gödel's theorems.

1.1 Main Results

1. The **Alignment Function**: Moral truth as set membership in Ω_{root} (§2).
2. The **Gödelian Completion**: Necessity of an external root authority (§3).
3. The **Cybernetic Stability Loop**: Systemic convergence under $S = H + P$ (§4).
4. The **Error Taxonomy**: Classification of ethical failure modes as computational errors (§5).

2 The Alignment Function

Definition 2.1 (Master Specification)**.** Let Ω_{root} be a consistent, complete specification of the optimal state of a universal system. An action A_i is *aligned* (morally correct) if and only if:

$$f(A_i) = 1 \iff A_i \subseteq \Omega_{\text{root}}. \tag{1}$$

[Geometric Morality] Moral truth is a rigid geometric property of the universal state-machine, not a subjective preference. The alignment function f is deterministic and binary: an action either conforms to Ω_{root} or it does not.

Definition 2.2 (Computational Sin)**.** An action A_i with $f(A_i) = 0$ is a *computational error*—it attempts to execute outside the allocated address space of Ω_{root}. We classify errors as:

[label=(ii)]

1. **Memory Leak:** Resources consumed without productive output.
2. **Segmentation Fault:** Access to forbidden address space.
3. **Stack Overflow:** Recursive self-reference without termination.

3 The Gödelian Completion

Theorem 3.1 (Necessity of External Root)**.** *Let $\mathcal{S}$ be any formal system sufficiently powerful to encode arithmetic (and therefore physics, ethics, or any domain with comparable expressive power). By Gödel's First Incompleteness Theorem [Gödel, 1931]:*

[label=(a)]

1. *$\mathcal{S}$ contains true statements that cannot be proved within $\mathcal{S}$.*
2. *The consistency of $\mathcal{S}$ cannot be established from within $\mathcal{S}$ (Second Incompleteness Theorem).*

Therefore, $\mathcal{S}$ requires an external oracle*—a root authority not contained within $\mathcal{S}$—to resolve undecidable propositions and verify its own consistency.*

[The Theological Mapping] This is a formal restatement of the classical cosmological argument: a self-contained system cannot ground its own existence. The external oracle in Gödel's framework corresponds structurally to the concept of a transcendent ground in classical theology.

4 The Cybernetic Stability Loop

Definition 4.1 (System Stability)**.** The stability S of a system at time t evolves according to:

$$S_{t+1} = S_t + \alpha(H - S_t) + \beta P, \tag{2}$$

where H is the Healing function (repair of accumulated damage), P is the Peace function (reduction of current environmental noise), and $\alpha, \beta > 0$ are gain coefficients representing the agent's "will" to execute repair.

Theorem 4.2 (Convergence to Stability)**.** *If $H > 0$ and $P > 0$ remain non-zero, the system asymptotically converges: $\lim_{t\to\infty} S_t = S^*$, where S^* is the fixed point satisfying $S^* = \alpha H/(1 + \alpha - 1) + \beta P$. If $H = 0$ (no healing), the system drifts into oscillatory chaos.*

5 Error Taxonomy

Computational Error	**Ethical Mapping**	**Correction**
Memory Leak	Acedia / Sloth	Garbage collection (repentance)
Segmentation Fault	Transgression	Access control (conscience)
Stack Overflow	Obsession / Addiction	Recursion limit (humility)
Race Condition	Injustice	Mutex lock (justice protocols)

6 Discussion

This paper inaugurates a research program in Computational Theology. Companion papers develop specific applications:

1. **Consciousness as Recursive Self-Reference** — Formalizes the observer as a Gödelian oracle.

2. **Free Will as Computational Necessity** — Proves agency is required for system novelty.

3. **Theodicy as Error Handling** — Reframes the Problem of Evil as fault tolerance in distributed systems.

6.1 Limitations

1. The alignment function f assumes the existence and consistency of Ω_{root}—this is an axiom, not a derivation.

2. The Gödelian argument establishes *structural necessity* of an external ground, not its specific nature or attributes.

3. The cybernetic model is a first approximation; real ethical systems involve higher-order feedback loops.

7 Conclusion

Computational Theology provides a formal language for theological reasoning that is consistent, falsifiable within its axioms, and interoperable with the existing tools of mathematics and computer science. The framework demonstrates that ethical truths have geometric structure, that self-contained systems require external grounding, and that systemic stability is achievable through the sustained application of healing and peace.

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